

Performance Testing

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Pytest + FastAPI TestClient for automated endpoint smoke tests; manual browser/video walkthrough for UI flow.
Type of Testing	Functional smoke testing and basic response-time sampling; not a formal stress/load test.
Target Module / API	/ and /health automated; /generate-workout and /submit-feedback are covered by manual/demo flow because they require Gemini generation.
Test Environment	Local Python environment using the submitted FitBuddy project.
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration	Expected Outcome
1	Open home page	1	Repeated smoke calls	Home page returns 200
2	Health endpoint	1	Repeated smoke calls	JSON status is ok
3	Submit valid profile	1	Manual / demo flow	Workout plan generated and stored
4	Submit feedback	1	Manual / demo flow	Updated plan generated and stored

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	Not formally benchmarked; basic local smoke requests used	N/A	AI generation latency depends on external Gemini API.
2	Response Time (Max)	< 5 seconds	Not formally benchmarked	N/A	No load test was executed.
3	Throughput	Recorded in load test	Not measured	N/A	Current submission is an educational local prototype.
4	Error Rate	< 1%	No repeated-load benchmark	N/A	Automated smoke tests are included; live Gemini generation still requires the declared google-genai dependency and an API key.
5	CPU Utilization	< 80%	Not measured	N/A	No profiler attached.
6	Memory Utilization	< 80%	Not measured	N/A	No profiler attached.

Step 4: Observations & Analysis

The project includes automated checks for the home and health endpoints. The AI-driven generation path depends on network access and a valid Gemini API key, so this submission does not invent response-time figures for the external service. The most relevant future performance work is load testing with realistic concurrency after deployment.

Step 5: Screenshots / Evidence

The uploaded demo video is included in the Project Demonstration folder. Representative frames are included as evidence of the development/demo flow.